Q1. Write a C++ program to evaluate the following expressions:

X=(-b-(b2-4ac))/2a

```
#include<iostream.h>
#include<conio.h>
void main(){
int x,a,b,c;
a=2;
b=3;
c=4;
clrscr();
x = (-b-(b*b-4*a*c))/2*a;
cout<<"\nX = "<<x;
getch();
}
```

Q2. Write a program which display the percentage of students which accept marks of three subjects from user.(Show the use of Implicit type casting)

```
#include<iostream.h>
#include<conio.h>
void main(){
int m1,m2,m3;
float per;
clrscr();
cout<<"\nEnter marks of 1st subject (100):- ";
cin>>m1;
cout<<"\nEnter marks of 2nd subject (100) :- ";
cin>>m2;
cout<<"\nEnter marks of 3rd subject (100) :- ";
cin>>m3;
per=(float)(m1+m2+m3)/300*100;
cout<<"\n\nPercentage of student :- "<<per;
getch();
}
```

Q3. Write a program to find area of circle such that the class circle must have three functions namely:
a)read() to accept the radius from user.
B)compute() for calculating the area
c)display() for displaying the result.(Use Scope resolution operator)

```
#include<iostream.h>
#include<conio.h>
#define PI 3.14
class Circle{
public:
int r;
float area;
void read();
void compute();
void display();
};
```

```
void Circle::read(){
cout<<"\nEnter the radius of the circle :- ";
cin>>r;
}
void Circle::compute(){
area=PI*r*r;
}
void Circle::display(){
cout<<"\nArea of Circle :- "<<area;
}
void main(){
Circle c;
clrscr();
c.read();
c.compute();
c.display();
getch();
}
```

Q4. Write a C++ program to calculate area of Rectangle using Inline function.

```
#include<iostream.h>
#include<conio.h>
float inline area(float length,float breadth){
return length*breadth;
}
void main(){
int l,b;
clrscr();
cout<<"\nEnter the length :- ";
cin>>l;
cout<<"\nEnter the breadth :- ";
cin>>b;
cout<<"\n\nArea of Rectangle :- "<<area(l,b);
getch();
}
```

Q5. WAP to declare a class calculation. Display addition, subtraction, multiplication, division of two numbers. Use friend function.

```
#include<iostream.h>
#include<conio.h>
class Calculation{
int a,b;
public:
Calculation(int x,int y){
a=x;
b=y;
}
friend void add(Calculation c);
friend void sub(Calculation c);
friend void mul(Calculation c);
friend void div(Calculation c);
```

```
};
void add(Calculation c){
cout<<"\nAddition :- "<<(c.a+c.b);
}
void sub(Calculation c){
cout<<"\nSubtraction :- "<<(c.a-c.b);
}
void mul(Calculation c){
cout<<"\nMultiplication :- "<<(c.a*c.b);
}
void div(Calculation c){
if(c.b!=0){
cout<<"\nDivision :- "<<(c.a/c.b);
}
else{
cout<<"\nCannot divide a number by zero!";
}
}
void main(){
int x,y;
clrscr();
cout<<"\nEnter the value of x :- ";
cin>>x;
cout<<"\nEnter the value of y :- ";
cin>>y;
Calculation c(x,y);
add(c);
sub(c);
mul(c);
div(c);
getch();
}
```

Q6. Write a C++ program to declare class 'Account' having data members as Account_No and Balance. Accept this data for 10 accounts and display data of accounts having balance greater than 10000.

```
#include<iostream.h>
#include<conio.h>
class Account{
public:
int acc_no;
int acc_bal;
void accept(){
cout<<"\nEnter Account No. :- ";
cin>>acc_no;
cout<<"\nEnter Account Balance :- ";
cin>>acc_bal;
}
void display(){
if(acc_bal>10000){
cout<<"\nAccount No. :- "<<acc_no;
cout<<"\nAccount Balance :- "<<acc_bal;
```

```
}
}
};
void main(){
Account a[10];  //Array of object
int i;
clrscr();
for(i=0;i<10;i++){
a[i].accept();
}
clrscr();
for(i=0;i<10;i++){
a[i].display();
}
getch();
}
```

Q7. WAP to declare class time having data members as hrs, min,sec. Write a constructor to accept data and display for two objects.

```
#include<iostream.h>
#include<conio.h>
class Time{
public:
int hrs,min,sec;
Time(int h,int m,int s){
hrs=h;
min=m;
sec=s;
}
void display(){
cout<<"\nHours :- "<<hrs;
cout<<"\nMinutes :- "<<min;
cout<<"\nSeconds :- "<<sec;
}
};
void main(){
clrscr();
Time t1(10,30,55);
t1.display();
getch();
}
```

Q8. Write a C++ program to define a class "Student" having data members roll_no, name. Derive a class "Marks" from "Student" having data members ml,m2,m3, total and percentage. Accept and display data for one student.

```
#include<iostream.h>
#include<conio.h>
class Student{
public:
int roll_no;
char name[30];
```

```
};
class Marks:public Student{
public:
int m1,m2,m3,total,per;
void accept(){
cout<<"\nEnter your Roll no. :- ";
cin>>roll_no;
cout<<"\nEnter your Name :- ";
cin>>name;
cout<<"\nEnter marks of 1st Subject :- ";
cin>>m1;
cout<<"\nEnter marks of 2nd Subject :- ";
cin>>m2;
cout<<"\nEnter marks of 3rd Subject :- ";
cin>>m3;
}
void display(){
cout<<"\nRoll no. :- "<<roll_no;
cout<<"\nName :- "<<name;
cout<<"\n1st Subject marks :- "<<m1;
cout<<"\n2nd Subject marks :- "<<m2;
cout<<"\n3rd subject marks :- "<<m3;
total=m1+m2+m3;
cout<<"\nTotal marks :- "<<total;
per=total/3;
cout<<"\nPercentage :- "<<per<<" %";
}
};
void main(){
Marks m;
clrscr();
m.accept();
m.display();
getch();
}
```

Q9. Write a C++ program to calculate the area and perimeter of rectangles using concept of inheritance.

```
#include<iostream.h>
#include<conio.h>
class Area{
public:
void area();
};
class Perimeter{
public:
void perimeter();
};
class Rectangle:public Area,public Perimeter{
public:
int length,breadth;
void get_data(){
```

```
cout<<"\nEnter the Length of Rectangle :- ";
cin>>length;
cout<<"\nEnter the Breadth of Rectangle :- ";
cin>>breadth;
}
void display(){
cout<<"\nLength of Rectangle :- "<<length;
cout<<"\nBreadth of Rectangle :- "<<breadth;
}
void area(){
int area;
area=length*breadth;
cout<<"\nArea of Rectangle :- "<<area;
}
void perimeter(){
int perimeter;
perimeter=2*(length+breadth);
cout<<"\nPerimeter of Rectangle :- "<<perimeter;
}
};
void main(){
Rectangle r;
clrscr();
r.get_data();
r.display();
r.area();
r.perimeter();
getch();
}
```

Q10. Write a C++ program to implement given figure class hierarchy.

```
#include<iostream.h>
#include<conio.h>
class Staff{
public:
int code;
};
class Teacher:public Staff{
public:
char subject[40];
void t_accept(){
cout<<"\nEnter Teacher details :- ";
cout<<"\nEnter your Code :- ";
cin>>code;
cout<<"\nEnter your Subject :- ";
cin>>subject;
}
void t_display(){
cout<<"\nTeacher details are :- ";
cout<<"\nCode :- "<<code;
cout<<"\nSubject :- "<<subject;
```

```
}
};
class Officer:public Staff{
public:
char grade;
void o_accept(){
cout<<"\nEnter Officer details :- ";
cout<<"\nEnter your Code :- ";
cin>>code;
cout<<"\nEnter your Grade :- ";
cin>>grade;
}
void o_display(){
cout<<"\nOfficer details are :- ";
cout<<"\nCode :- "<<code;
cout<<"\nGrade :- "<<grade;
}
};
void main(){
Teacher t1;
Officer o1;
clrscr();
t1.t_accept();
t1.t_display();
getch();
clrscr();
o1.o_accept();
o1.o_display();
getch();
}
```

Q11. Write a C++ program to implement given figure class hierarchy. Assume suitable member variables and member functions.

```
#include<iostream.h>
#include<conio.h>
class Account{
public:
char name[30];
int acc_no;
};
class Saving_acc:virtual public Account{
public:
int saving_bal;
};
class Current_acc:virtual public Account{
public:
int current_bal;
};
class Fix_depo_acc:public Saving_acc,public Current_acc{
public:
int fix_bal;
```

```
void accept(){
cout<<"\nEnter your Name :- ";
cin>>name;
cout<<"\nEnter your Account no. :- ";
cin>>acc_no;
cout<<"\nEnter your Saving balance :- ";
cin>>saving_bal;
cout<<"\nEnter your Current balance :- ";
cin>>current_bal;
cout<<"\nEnter your Fixed Deposit balance :- ";
cin>>fix_bal;
}
void display(){
cout<<"\nName :- "<<name;
cout<<"\nAccount no. :- "<<acc_no;
cout<<"\nSaving balance :- "<<saving_bal;
cout<<"\nCurrent balance :- "<<current_bal;
cout<<"\nFixed Deposit balance :- "<<fix_bal;
}
};
void main(){
Fix_depo_acc f;
clrscr();
f.accept();
f.display();
getch();
}
```

Q12. Write a program which show the use of constructor in derived class.

```
#include<iostream.h>
#include<conio.h>
class Base{
int x;
public:
Base(int a){
x=a;
cout<<"\nConstructor in Base class :- "<<x;
}
};
class Derived:public Base{
int y;
public:
Derived(int a,int b):Base(a){
y=b;
cout<<"\nConstructor in Derived class :- "<<y;
}
};
void main(){
clrscr();
Derived(2,3);
getch();
```

```
}
```

Q13. Write a C++ program to declare a class “Book” containing data members Book_name, auther_name, and price . Accept this information for one object of the class using pointer to that object.

```
#include<iostream.h>
#include<conio.h>
class Book{
public:
char book_name[40];
char author_name[40];
int price;
void accept(){
cout<<"\nEnter Book details :- ";
cout<<"\nEnter Book Name :- ";
cin>>book_name;
cout<<"\nEnter Author Name :- ";
cin>>author_name;
cout<<"\nEnter Book's Price :- ";
cin>>price;
}
void display(){
cout<<"\nBook details are :- ";
cout<<"\nBook Name :- "<<book_name;
cout<<"\nAuthor Name :- "<<author_name;
cout<<"\nPrice :- "<<price;
}
};
void main(){
Book b,*ptr;
clrscr();
ptr=&b;
ptr->accept();
ptr->display();
getch();
}
```

Q14. Write a C++ program to declare a class "Box" having data members height, width and breadth. Accept this information for one object using pointer tothat object. Display the area and volume of that object.

```
#include<iostream.h>
#include<conio.h>
class Box{
public:
int height,width,breadth;
void accept(){
cout<<"\nEnter Height of box :- ";
cin>>height;
cout<<"\nEnter Width of box :- ";
cin>>width;
cout<<"\nEnter Breadth of box :- ";
cin>>breadth;
```

```
}
void display(){
int area,volume;
area=2*(height+width+breadth);
cout<<"\nArea of box :- "<<area;
volume=height*width*breadth;
cout<<"\nVolume of box :- "<<volume;
}
};
void main(){
Box b,*ptr;
clrscr();
ptr=&b;
ptr->accept();
ptr->display();
getch();
}
```

Q15. Write a C++ program declare a class "polygon" having data members width and height. Derive classes "rectangle" and "triangle" from "polygon" havingArea() as a member function. Calculate area of triangle and rectangle usingPointer to derived class object.

```
#include<iostream.h>
#include<conio.h>
class Polygon{
public:
int width,height;
void accept(int w,int h){
width=w;
height=h;
}
virtual float area(){
return 0;
}
};
class Rectangle:public Polygon{
public:
float area(){
return width*height;
}
};
class Triangle:public Polygon{
public:
float area(){
return (width*height)/2.0;
}
};
void main(){
Rectangle r;
Triangle t;
Polygon *ptr;
clrscr();
```

```
ptr=&r;
ptr->accept(10,5);
cout<<"\nArea of Rectangle :- "<<ptr->area();
ptr=&t;
ptr->accept(10,5);
cout<<"\nArea of Triangle :- "<<ptr->area();
getch();
}
```

Q16. Write a C++ Program to interchange the values of two int , float and Char using function overloading.

```
#include<iostream.h>
#include<conio.h>
void swap(int a,int b){
int temp;
cout<<"\nBefore swapping :- ";
cout<<"\n"<<a<<"\n"<<b;
temp=a;
a=b;
b=temp;
cout<<"\nAfter swapping :- ";
cout<<"\n"<<a<<"\n"<<b;
}
void swap(float a,float b){
float temp;
cout<<"\nBefore swapping :- ";
cout<<"\n"<<a<<"\n"<<b;
temp=a;
a=b;
b=temp;
cout<<"\nAfter swapping :- ";
cout<<"\n"<<a<<"\n"<<b;
}
void swap(char a,char b){
char temp;
cout<<"\nBefore swapping :- ";
cout<<"\n"<<a<<"\n"<<b;
temp=a;
a=b;
b=temp;
cout<<"\nAfter swapping :- ";
cout<<"\n"<<a<<"\n"<<b;
}
void main(){
clrscr();
swap(10,20);
swap(1.2f,3.4f);
swap('x','y');
getch();
}
```

Q17. Write a C++ Program that find the sum of two int and double number using function overloading.

```
#include<iostream.h>
#include<conio.h>
void sum(int a,int b){
int result;
cout<<"\n\nValue of a :- "<<a;
cout<<"\nValue of b :- "<<b;
result=a+b;
cout<<"\nSum of integers :- "<<result;
}
void sum(double x,double y){
double result;
cout<<"\n\nValue of x :- "<<x;
cout<<"\nValue of y :- "<<y;
result=x+y;
cout<<"\nSum of doubles :- "<<result;
}
void main(){
clrscr();
sum(10,20);
sum(1.2345,6.7890);
getch();
}
```

Q18. Define a class parent in which use read function, define another class child use same read function. Display the data of both the read function on output screen using virtual function.

```
#include<iostream.h>
#include<conio.h>
class Parent{
public:
int a;
virtual void read(){
cout<<"\nEnter value for Parent(a) :- ";
cin>>a;
}
virtual void display(){
cout<<"\nParent data(a) :- "<<a;
}
};
class Child:public Parent{
public:
int b;
void read(){
cout<<"\nEnter value for Child(b) :- ";
cin>>b;
}
void display(){
cout<<"\nChild data(b) :- "<<b;
}
};
void main(){
```

```
Parent p,*ptr;
Child c;
clrscr();
ptr=&p;
ptr->read();
ptr->display();
ptr=&c;
ptr->read();
ptr->display();
getch();
}
```

Q19. Write a program which show the use of Abstract class.

```
#include<iostream.h>
#include<conio.h>
class Shape{
public:
virtual void draw() = 0;
};
class Circle:public Shape{
public:
void draw(){
cout<<"\nDrawing Circle ";
}
};
void main(){
Circle c;
clrscr();
c.draw();
getch();
}
```